

Bayesian Fraud Detection

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1 Introduction

Modern financial systems generate large volumes of transactional data that evolve over time. A key challenge in this context is the detection of anomalous or potentially fraudulent transactions.

This report addresses the problem of anomaly detection in financial transaction data from a Bayesian perspective.

My personal Master Thesis project develops a structured modeling framework based on Relational Event Models (REMs) and latent variables for considering anomalies, while in this case we consider a different viewpoint, based on an effective distribution of the estimated parameters.

Instead of explicitly modeling the interaction structure between entities and the temporal evolution of events, we directly model the probability that a transaction is anomalous. This leads to a different probabilistic formulation that focuses on inference rather than on the full generative dynamics of the system.

Within this setting, Bayesian methods provide a natural alternative to more complex event-based models. We first illustrate the potential of Bayesian inference through a conjugate Beta-Binomial model, which provides a simple and analytically tractable posterior distribution for the global anomaly rate. This first step shows how Bayesian methods can describe uncertainty about the distribution of anomalies rather than returning only a fixed estimate.

We then extend the analysis to Bayesian logistic regression, where transaction-level covariates are included and the model estimates a posterior distribution over the regression parameters using Markov Chain Monte Carlo (MCMC). In this way, the focus moves from point estimation to full posterior inference, allowing uncertainty in the model parameters to be propagated into anomaly predictions.

The purpose of this work is to show how Bayesian inference can be used as an alternative, more compact approach to anomaly detection.

2 Data Description

The dataset used in this work is publicly available on the Kaggle bank transaction dataset by Watsky and contains anonymous bank transaction records. Each observation corresponds to a single financial transaction over time, recorded in a banking system for a single account holder. Unlike relational transaction datasets, where interactions occur between a sender and a receiver, this dataset does not contain explicit dyadic structure. Instead, it represents a uni-variate sequence of financial events associated with one entity evolving over time, where each transaction affects the account balance.

The raw dataset contains 116201 transactions in total, and includes variables such as transaction date, withdrawal amount, deposit amount, cheque number, and account balance. In the preprocessing phase, structurally irrelevant variables (e.g. cheque number and textual descriptions) were removed.

Since withdrawal and deposit represent mutually exclusive outcomes of the same transaction, they were merged into a single monetary variable to obtain a consistent representation of financial flow. No observations were removed during preprocessing; only non-informative or redundant variables were excluded in order to preserve the original transactional structure of the dataset.

Variable	Description
amount	Unified transaction value derived from withdrawal/deposit
balance	Account balance after or before the transaction
date	Timestamp of the transaction
time index	Time from the first occurred event
log time	Log-transformed time index
is withdrawal	Binary indicator for withdrawal transactions
balance amount ratio	Ratio between balance and absolute transaction amount
weekday	Day of the week of the transaction
Y	Binary anomaly label obtained via Isolation Forest

Table 1: Summary of variables used in the analysis

3 Anomaly Label Construction

Since true fraud labels are not available, for the first part of the project we construct a proxy variable Y_i using an unsupervised anomaly detection model.

We employ Isolation Forest algorithm proposed by Liu et al. (2008) to assign anomaly scores to all transactions. In particular, a transaction is labeled as anomalous if its score exceeds the 95th percentile:

$$Y_i = \begin{cases} 1 & \text{if anomaly} \\ 0 & \text{otherwise} \end{cases}$$

This produces a binary indicator used in the Bayesian models. The Isolation Forest procedure identifies 5810 anomalies, corresponding to an anomaly rate of 5.000%. This proportion is a direct consequence of the labeling rule: by defining anomalies as the

observations above the 95% quantile of the Isolation Forest score distribution, the anomaly share is fixed at approximately 5% by construction. Therefore, the global anomaly rate of 5% should be interpreted as a modeling precondition rather than as an independently estimated empirical result.

This global precondition will be used in the next section as prior information for the Bayesian Beta-Binomial model. In particular, when testing the model on a random subsample, the prior distribution can encode the assumption that the expected anomaly probability is close to 5%.

Quantity	Value
Total observations	116201
Detected anomalies	5810
Anomaly rate	0.0500
95% score threshold	0.4765
Minimum anomaly score	0.3563
Mean anomaly score	0.3997
Maximum anomaly score	0.7816

Table 2: Isolation Forest anomaly label summary

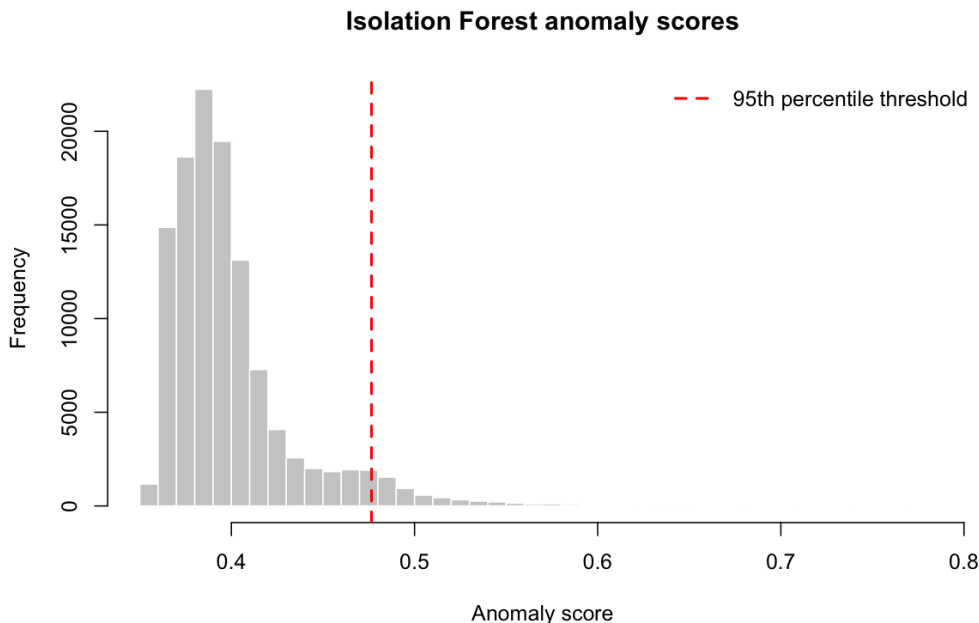


Figure 1: Distribution of Isolation Forest anomaly scores. The dashed line marks the 95th percentile threshold.

4 Bayesian Beta-Binomial Model

We now model the global anomaly rate π as $Y_i \sim \text{Bernoulli}(\pi)$, with prior parameter distribution $\pi \sim \text{Beta}(\alpha_0, \beta_0)$. The prior is chosen to reflect the global precondition

introduced in the anomaly-label construction step, namely that approximately 5% of transactions are treated as anomalous.

This choice is motivated by the conjugacy between the Beta prior and the Bernoulli likelihood. Conjugacy implies that the posterior distribution belongs to the same family as the prior, leading to a closed-form update:

$$\pi \mid Y \sim \text{Beta}(\alpha_0 + k, \beta_0 + n - k),$$

where $k = \sum_{i=1}^n y_i$ is the number of observed anomalies of n observations.

4.1 Interpretation

The posterior expectation value with $\alpha = \alpha_0 + k$ and $\beta = \beta_0 + n - k$ is so:

$$\mathbb{E}[\pi \mid Y] = \frac{\alpha}{\alpha + \beta}.$$

The Beta-Binomial analysis is performed on a random subsample of the original dataset of 250 transactions, in which 13 anomalies are observed. This corresponds to an empirical anomaly rate of 5.2% in the subsample.

Prior	α	β	Posterior $\mathbb{E}$	95% CI low	95% CI high
Beta(1,1)	14	238	0.0556	0.0308	0.0869
Beta(1,19)	14	256	0.0519	0.0287	0.0812
Beta(5,95)	18	332	0.0514	0.0309	0.0769

Table 3: Posterior estimates under different Beta priors

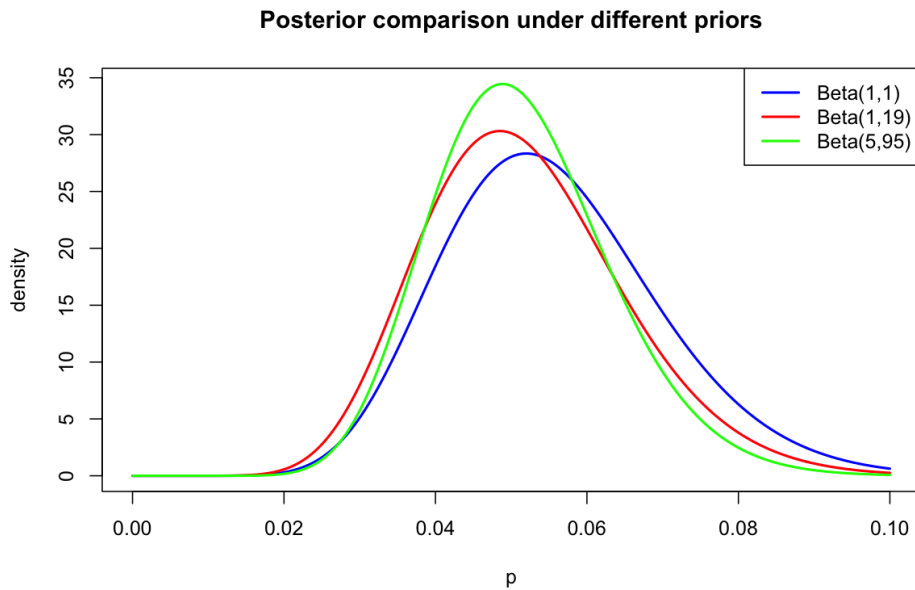


Figure 2: Posterior distributions of anomaly probability under three different Beta priors.

As the prior becomes more informative, the posterior expectation progressively shifts toward the prior belief of an overall anomaly rate close to 5%. In particular, stronger priors reduce the influence of the sampled observations, leading to posterior estimates that are more concentrated around the prior information and exhibit more concentrated credible intervals.

5 Sequential Bayesian Learning

To analyze how the anomaly probability evolves over time, we consider an online update scheme. In this section, the initial parameters α_0 and β_0 are set to represent a non-informative prior. In practice, this corresponds to the case where no strong prior belief is imposed before observing the sequence of transactions, so the posterior anomaly rate is mainly driven by the data accumulated over time.

At each time step t , we update:

$$\alpha_t = \alpha_0 + k_t, \quad \beta_t = \beta_0 + t - k_t,$$

where k_t is the cumulative number of anomalies observed up to time t .

The posterior expectation evolves as:

$$\mathbb{E}[\pi_t | Y_t] = \frac{\alpha_t}{\alpha_t + \beta_t}.$$

This allows us to track how the estimated anomaly probability changes as more transactions are observed. As the number of observed transactions increases, the posterior credible interval becomes progressively narrower, because the model accumulates more evidence about the global anomaly rate. Therefore, the sequential procedure shows not only the evolution of the posterior mean, but also how the uncertainty around the anomaly-rate estimate decreases over time.

At the end of the sequential updating procedure, the posterior mean anomaly probability is 0.050007. The final 95% credible interval is $[0.048960, 0.051063]$, showing that after observing the full sequence the global anomaly rate is estimated very precisely.

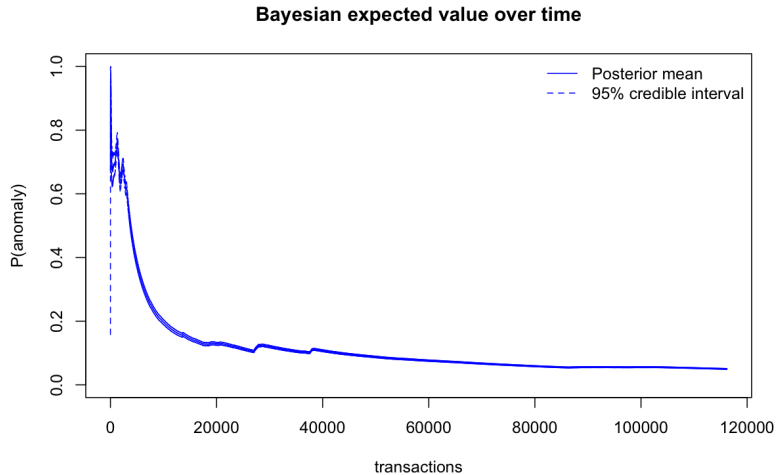


Figure 3: Sequential posterior mean of the anomaly probability with 95% credible interval.

6 Bayesian Logistic Regression via MCMC

While the Beta-Binomial model provides a global estimate of the anomaly rate, it does not incorporate transaction-level information. To model how transaction characteristics influence the probability of anomalous behaviour, we extend the analysis through Bayesian logistic regression.

Given a binary anomaly indicator $Y_i \in \{0, 1\}$ for transaction i , we assume $Y_i \sim \text{Bernoulli}(\pi_i)$ where π_i represents the probability that transaction i is anomalous. The probability is linked to the observed covariates through a logistic regression model:

$$\text{logit}(\pi_i) = \log \frac{\pi_i}{1 - \pi_i} = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip}.$$

The predictors include standardized transaction amount, standardized balance, standardized log-time, withdrawal indicator, balance-to-amount ratio, and weekday effects.

Differently from the Beta-Binomial setting, the posterior distribution of the regression coefficients cannot be derived analytically due to the non-conjugate logistic likelihood. For this reason, posterior inference is performed through Markov Chain Monte Carlo (MCMC) methods.

6.1 MCMC intuition

Instead of producing a single point estimate for the parameters, MCMC generates a sequence of samples:

$$\beta^{(1)}, \beta^{(2)}, \dots, \beta^{(s)} \sim p(\beta \mid Y, X),$$

which approximate the posterior distribution of the model parameters. Each draw corresponds to one plausible configuration of the regression coefficients given the observed data.

More specifically, each sampled vector contains one coefficient for every feature included in the model $\beta^{(s)} = (\beta_0^{(s)}, \beta_1^{(s)}, \dots, \beta_p^{(s)})$, where each component represents the effect of a specific transaction feature on the anomaly probability.

The sampling procedure constructs a Markov chain whose stationary distribution coincides with the posterior distribution of the parameters. After an initial warm-up phase, the chain progressively explores the high-probability regions of the posterior space. The objective is to obtain samples that are as weakly correlated as possible, so that successive draws behave approximately as independent realizations from the target posterior distribution.

Predictions are therefore obtained by averaging over many sampled parameter configurations rather than relying on a single estimated model. This allows the Bayesian framework to naturally quantify predictive uncertainty.

In particular, for each transaction the model provides a posterior mean probability of fraud and a credible interval describing uncertainty around the prediction.

This represents one of the main conceptual differences with respect to the frequentist logistic regression approach, which produces only point estimates of the probabilities.

The Bayesian logistic regression was fitted on 92960 training observations and evaluated on 23241 test observations. The training anomaly rate is 0.050022, while the test anomaly rate is 0.049912.

Three prior specifications of all β_j parameters were compared for the regression coefficients and intercept: a standard normal prior $\mathcal{N}(0, 1)$, a weakly informative normal prior $\mathcal{N}(0, 2)$, and a Cauchy(0, 2.5) prior. The Cauchy(0, 2.5) prior follows the recommendation of Gelman et al. (2008), which is commonly used as a weakly informative default for Bayesian logistic regression.

Prior	Max $\hat{R}$	Min bulk ESS	Min tail ESS
Normal(0,1)	1.003	1045.7	1251
Normal(0,2)	1.005	831.1	1318
Cauchy(0,2.5)	1.003	807.8	1150

Table 4: MCMC diagnostic summary

The convergence diagnostics are satisfactory: all fixed-effect $\hat{R}$ values are close to 1, and the effective sample sizes are sufficiently large for the fixed effects.

The regression coefficients can be interpreted on the log-odds scale. For the Cauchy(0, 2.5) model, the coefficient of *amount* is positive (2.05), meaning that larger standardized transaction amounts increase the posterior probability of being classified as anomalous. In odds terms, a one-standard-deviation increase in amount multiplies the odds by approximately $\exp(2.05) \approx 7.8$, holding the other variables fixed. By contrast, *log_time* has a negative coefficient (-1.84), so later transactions in the sequence are associated with lower anomaly odds, approximately multiplied by $\exp(-1.84) \approx 0.16$. The variable *balance_amount_ratio* has a strongly negative coefficient, suggesting that transactions that are large relative to the balance are much more likely to be anomalous.

7 Comparison of Bayesian Approaches

The two Bayesian approaches considered in this work address anomaly detection at different levels of complexity.

The Beta-Binomial model provides a simple probabilistic description of the overall anomaly frequency in the dataset. Due to the conjugacy between the Beta prior and the Bernoulli likelihood, posterior inference admits a closed-form solution and can be computed analytically. This makes the model highly interpretable and computationally efficient, although limited to estimating a global anomaly probability.

The Bayesian logistic regression model extends the analysis by incorporating transaction-specific covariates. In this case, posterior inference no longer admits a closed-form expression and requires simulation-based methods such as MCMC. Although computationally more expensive, this approach allows the anomaly probability to vary across transactions according to their observed characteristics.

From a predictive perspective, all Bayesian logistic regressions outperform the frequentist logistic regression in ROC-AUC, log loss, and F1-score when using the fixed threshold 0.5.

The final Bayesian row in the performance table uses the Cauchy(0, 2.5) model, which gives the best metrics. The only difference is the classification threshold: with the standard threshold 0.5, the model has precision 0.7849 and recall 0.6198. With the F1-optimized threshold 0.1493, more observations are classified as anomalies, increasing recall to 0.9172 and F1-score to 0.7160, but reducing precision to 0.5872.

Thus, the F1-optimized threshold gives a better balance between precision and recall than the fixed threshold 0.5. This is particularly relevant in fraud detection, where maximizing recall (i.e. catching as many fraudulent transactions as possible) is often more important than maximizing precision (i.e. minimizing false positives), since the cost of missing a fraudulent transaction can be much higher than the cost of flagging a legitimate transaction.

Model	Threshold	AUC	Accuracy	Log loss	Recall	F1
Frequentist	0.50	0.9313	0.9670	0.0966	0.4707	0.5877
Bayes N(0,1)	0.50	0.9767	0.9725	0.0680	0.6103	0.6894
Bayes N(0,2)	0.50	0.9781	0.9725	0.0675	0.6164	0.6915
Bayes Cauchy(0,2.5)	0.50	0.9790	0.9725	0.0674	0.6198	0.6927
Bayes Cauchy F1-opt.	0.1493	0.9790	0.9637	0.0674	0.9172	0.7160

Table 5: Predictive performance on the test set

Model	TP	TN	FP	FN
Frequentist	546	21929	152	614
Bayes N(0,1)	708	21895	186	452
Bayes N(0,2)	715	21888	193	445
Bayes Cauchy(0,2.5)	719	21884	197	441
Bayes Cauchy F1-opt.	1064	21333	748	96

Table 6: Confusion matrix components on the test set

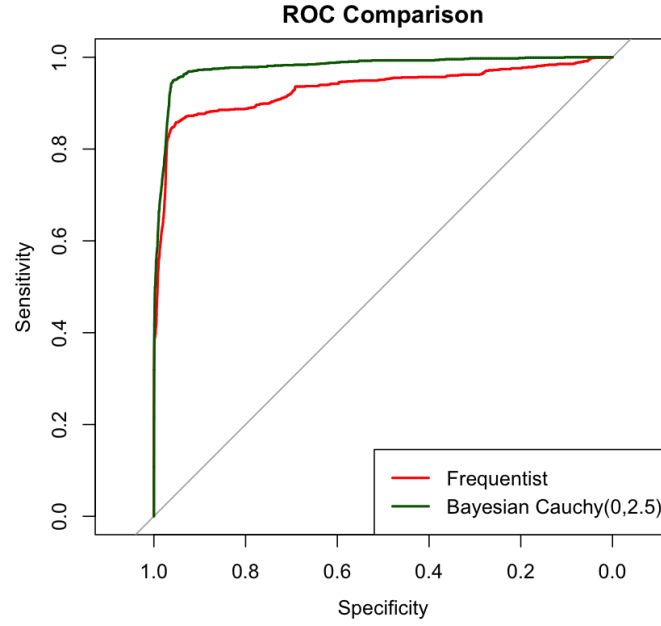


Figure 4: ROC curve comparison between frequentist logistic regression and the Bayesian logistic regression with $\text{Cauchy}(0, 2.5)$ prior.

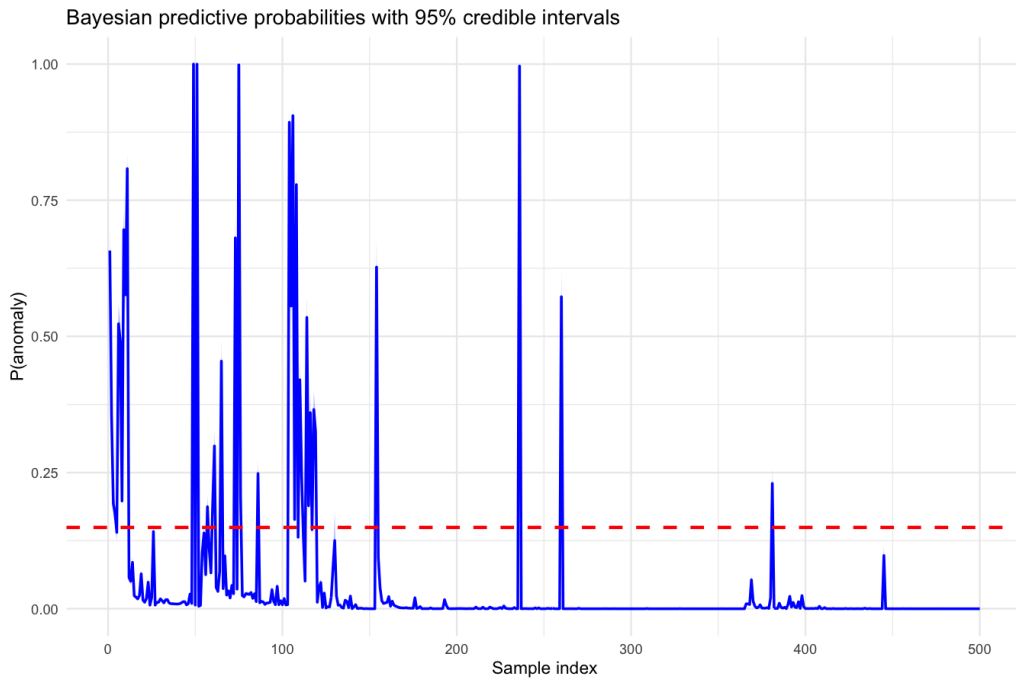


Figure 5: Bayesian posterior predictive probabilities with 95% credible intervals for a sample of test observations.

The Bayesian framework therefore provides not only point predictions and classification performance, but also a direct quantification of uncertainty. This additional uncertainty quantification is particularly relevant in fraud detection applications, where identifying uncertain or borderline cases may be as important as maximizing accuracy.

8 Discussion

The results show that Bayesian logistic regression is a strong and compact alternative for this anomaly-detection setting: it improves discrimination and calibration metrics relative to the frequentist baseline, while also providing posterior uncertainty. Among the tested priors, $\text{Cauchy}(0, 2.5)$ provides the best overall predictive trade-off, and its F1-optimized threshold further improves recall without collapsing precision.

At the same time, the analysis has structural limits that should be kept explicit. The target label is generated by Isolation Forest rather than a ground-truth fraud indicator, so model quality is measured against an unsupervised definition of anomaly. This means that high predictive performance should be interpreted as consistency with the proxy-labeling mechanism, not necessarily as direct fraud-detection accuracy in production.

From an operational viewpoint, the current framework is useful as a risk-scoring system: the posterior mean can drive prioritization, while credible intervals can flag uncertain cases for manual review. A natural next step is to evaluate stability over time with temporal splits and, where possible, validate against confirmed fraud outcomes.

Code Availability

The complete source code used for data preprocessing, model estimation, MCMC diagnostics, and figure generation is available at:

<https://github.com/giolagioia/Bayes-Anomaly-Detection>

References

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